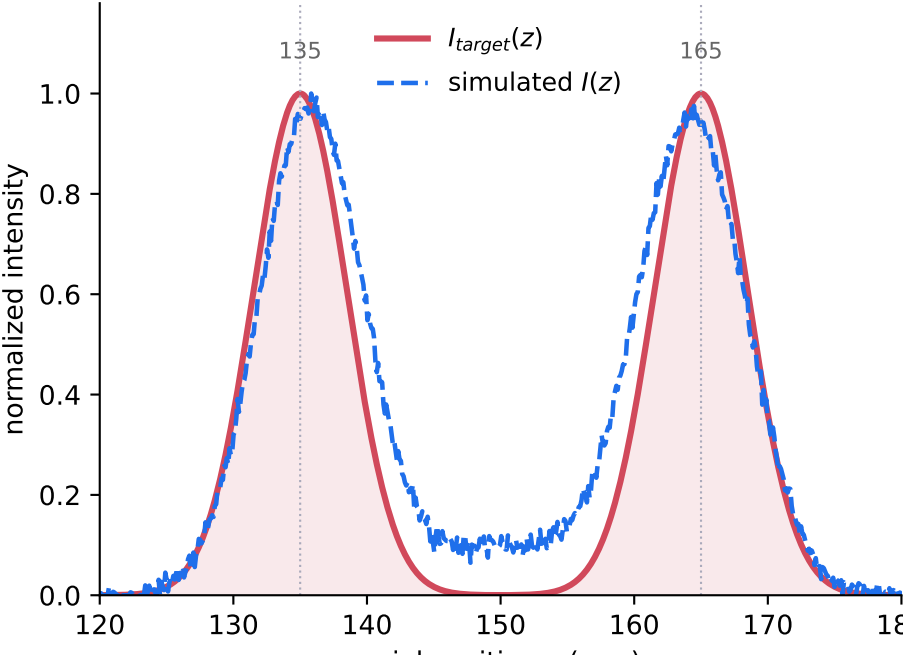


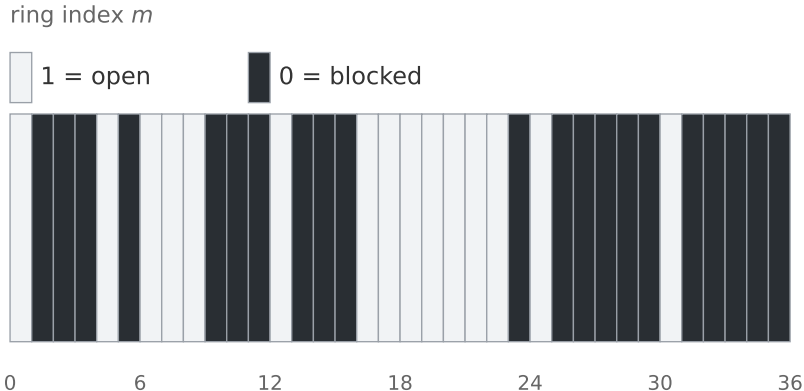
(a) Target vs. simulated axial intensity



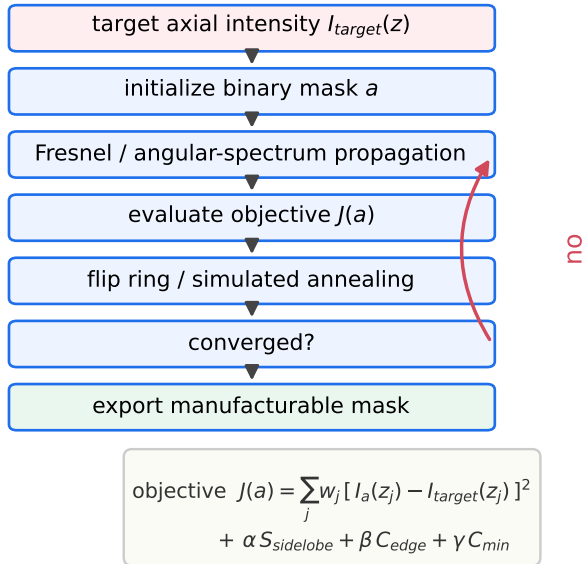
(c) Generated binary zone-plate mask



(b) Binary ring variables $a_m \in \{0, 1\}$



(d) Target-driven optimization loop



$$\text{objective } J(a) = \sum_j w_j [I_a(z_j) - I_{\text{target}}(z_j)]^2 + \alpha S_{\text{sidelobe}} + \beta C_{\text{edge}} + \gamma C_{\text{min}}$$